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Personal Reflection – Sprint 6

During this sprint, my focus was expanding the application beyond the quiz feature I worked on previously. I built the flashcard system so users can review vocabulary in a simple format, with words fetched from Firestore the same way as the quiz and library. This was a big step toward making learning more interactive and personal. I also implemented logic that preserves the current page when refreshing the browser, so users no longer get redirected to the login screen. This required modifying how user state was stored and handled client-side, and it helped improve the overall usability of the app.

I put a lot of effort into the profile page as well. I added avatar upload functionality, connected to Firebase storage, so users can upload their photo and see it instantly. I also integrated the XP and level system, where users gain experience from activities and progress toward new levels. To create more variety, I developed the "match words" activity, which allows users to match vocabulary terms to definitions. Finally, I enhanced the visual layout of saved words and made the UI more consistent across sections.

The biggest challenges I faced involved keeping the different features connected with the user's data. For example, ensuring profile updates were synced with Firestore, or making sure the XP system updated correctly after finishing an activity. I had to test different approaches to state management and fetch logic until everything behaved predictably. I also ran into issues when refreshing, because Firebase authentication state can take a moment to load, which initially caused redirection problems. I learned how to handle that by waiting for auth state resolution.

From this sprint I learned how important thoughtful user experience is. Each feature felt separate at first, but once I connected them to the user profile and stored progress, the app started to feel more cohesive. Next sprint, I want to improve planning before coding, and refactor places where logic became repetitive. This will help reduce bugs, make the project cleaner, and allow future features to be added more smoothly.

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